From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

FIITJEE

JEE(Main)-2014

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

CRT – I
(Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	D	A
2.	B	B	C
3.	C	C	C
4.	D	C	B
5.	D	C	A
6.	A	A	B
7.	C	B	C
8.	D	D	D
9.	B	B	A
10.	C	A	B
11.	C	C	C
12.	B	B	B
13.	D	B	B
14.	D	A	C
15.	A	B	B
16.	C	B	C
17.	B	B	A
18.	B	D	B
19.	B	B	C
20.	B	B	B
21.	C	C	C
22.	A	C	B
23.	D	D	A
24.	B	B	A
25.	B	C	B
26.	C	C	C
27.	D	D	C
28.	B	B	C
29.	C	C	B
30.	B	A	C

Physics

PART – I

1. $\rho = \dfrac{3\cdot14\times10^{-6}\times5}{1}$

$= 15.7\mu$ ohm metre.

$\delta\rho = \rho\left[\dfrac{2\delta r}{r} + \dfrac{\delta L}{L} + \dfrac{\delta R}{R}\right]$ ohm metre.

$= 15\cdot7\mu\left[2\times\dfrac{\cdot01}{1} + \dfrac{\cdot01}{1} + \dfrac{\cdot01}{5}\right]$ ohm metre

$= 15.7\mu\left[\cdot02 + \cdot01 + \cdot002\right]$ ohm – metre

$= 15.7\times\cdot032\mu\,\Omega$ m

$= \cdot5024\mu\,\Omega$ m

2. $100\times\dfrac{\Delta P}{P} = 3\times0\cdot1 + 2\times0\cdot2 + 3\times0\cdot3 + 4\times0\cdot1$

$= \cdot3 + \cdot4 + \cdot9 + \cdot4 = 2\%$

3. $A = \lambda N_0 e^{-\lambda t_0}$

$3Ae = 3\lambda N_0 e^{-\lambda t}$

$\Rightarrow \dfrac{1}{e} = e^{\lambda(t-t_0)} = e^{-1}$

$t = t_0 - \dfrac{1}{\lambda}.$

4. At $t = \dfrac{v_o \sin\theta}{g}, \vec{v_1} = v_0\cos\theta\hat{i}$

$\vec{v_2} = v_0\hat{k} - gt\hat{j}$

$\therefore \vec{v_2} - \vec{v_1} = v_0\hat{k} - gt\,\hat{j} - v_0\cos\theta\hat{i}$

5. $f = a^2\left(Z-b\right)^2.$

$\therefore$ (D)

6. $-v\dfrac{dv}{dx} = \dfrac{\left(k+\alpha mg\right)x}{m}$

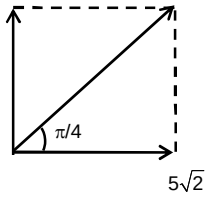
$\Rightarrow -\int_{v_0}^{0} vdv = \left(\dfrac{k+\alpha mg}{m}\right)\int_{o}^{x_0} xdx$

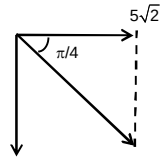
$\Rightarrow x_0 = v_0\sqrt{\dfrac{m}{\left(k+\alpha mg\right)}}$

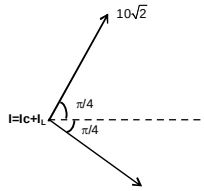
7. $\dfrac{I_{max}}{I_{min}} = \dfrac{\left(\sqrt{I_1}+\sqrt{I_2}\right)^2}{\left(\sqrt{I_1}-\sqrt{I_2}\right)^2} = \alpha$

$\frac{I_1}{I_2} = \left(\frac{\sqrt{\alpha}+1}{\sqrt{\alpha}-1}\right)^2$ if $I_1 > I_2$

8. $I_c = \frac{100}{5\sqrt{2}}\sin\left(50t + \frac{\pi}{4}\right)$







$I_L = \frac{100}{5\sqrt{2}}\sin\left[50t - \frac{\pi}{4}\right]$

$I = 20\sin(50t)$

9. $2mg\sin 30 - T = 2m(a)$

$T = ma$

$\Rightarrow a = g/3 \;\&\; T = \frac{mg}{3}$

$\therefore$ Net force =

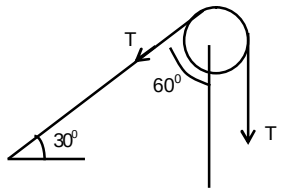
Result of these forces $= T\left(-\hat{j}\right) + T\cos 60^0\left(-\hat{j}\right) + T\sin 60\left(-\hat{i}\right)$

$= -\frac{\sqrt{3}}{2}T\hat{i} - \frac{3}{2}T\hat{j}$

$= \frac{mg}{6}\left(-\sqrt{3}\hat{i} - 3\hat{j}\right)$

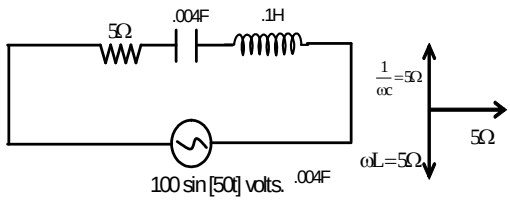
As net force in pulley = 0,

force by clamp $= \frac{mg}{6}\left(\sqrt{3}\hat{i} + 3\hat{j}\right)$



10. The circuit can be simplified as:

$\phi = 0^0$,peak current =20 Ampere.



12. When u decreases v increases. For plane mirror u decreases.

$\therefore$ the image in plane mirror moves towards the left.

13. $4 \times \frac{\mu_0 I}{4\pi}.\left[\sin 45^0 + \sin 45^0\right]\left[\frac{1}{R} - \frac{1}{2R}\right]$

$= \frac{\mu_0 I}{\pi}\left[\sqrt{2}\right]\left[\frac{1}{2R}\right] = \frac{\mu_0 I}{\sqrt{2}\pi R}$ $\odot$

14. $V_1 = \frac{10}{4+4} \times 4$ volts = 5 volt

$V_2 = \frac{10 \times \frac{100 \times 4}{100+4}}{4 + \frac{100 \times 4}{100+4}}$

= 4.9 volts $\therefore \left|V_1 - V_2\right| = 0.1$ volt

15. In both cases
applying com,
$mv_0 = Mv$

$\Rightarrow V = \frac{mv_0}{M}$

$\therefore \frac{V_1}{V_2} = 1:1$

16. In steady state :-

$V_P - V_Q = \frac{E}{2R} \times R = \frac{E}{2}$

$V_Q - V_S = \frac{E}{2R} \times R = \frac{E}{2}$

$V_S = V_R$

$V_P - V_R = E$

$\therefore Q_A = CE$

$Q_B = \frac{CE}{2}$

17. $V_P - V_0 = -[10 \times 10 + 10 \times 10]$
= - 200 volts.

18. For the two blocks to move together

$\mu_{min} = \frac{F}{(M+m)g} = \frac{F}{2Mg}$

As $\mu > \mu_{min}$,
the two blocks move together.
$\therefore$ Velocity of the system (M+m) is

$v = at = \left(\frac{F}{2M}\right)t$

$\Rightarrow P = Fv = F\frac{F}{2m}t = \frac{F^2}{2M}t.$

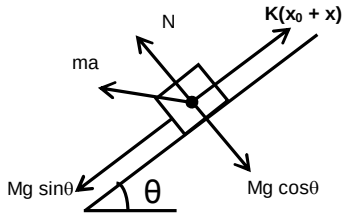
19. $\frac{P}{4\pi r^2} \times Axt = mcT$

$t = \frac{4\pi r^2 mcT}{PA}$

20. $\Delta A = \pi \ell b(2\alpha)T$.

21. After collision,
the rod starts rotating about its centre of mass.
Now the torque of mg about the centre of mass = 0, hence ω is constant.

22. x_0 ∴ elongation in equilibrium



$\therefore ma\cos\theta + mg\sin\theta = kx_0$ (1)

For x = elongation from equilibrium

net force $= mg\sin\theta + ma\cos\theta - k(x_0 + x)$

$= -kx$ (i)

$\therefore acc = -\frac{kx}{m}$

$\Rightarrow T = 2\pi\sqrt{\frac{m}{k}}$

23. $\frac{4}{3}\pi r^3[n - n_0]g = 6\pi\mu r V_T.$

$V_T = \frac{2r^2[n - n_0]}{9\mu}g$

$P_g = \frac{4}{3}\pi r^3.n.g.\frac{2r^2(n - n_0)g}{9\mu}$

$= \frac{8\pi r^5 n}{27\mu}[n - n_0]g^2$ 55. $\Delta x = 2cm = 0.02m$

24. $\Delta\phi = k\Delta x = 100 \times .02$
= 2rad

25. $P_0 - \frac{2T}{r} + \rho gh = P_0 - \frac{2T}{2r}$

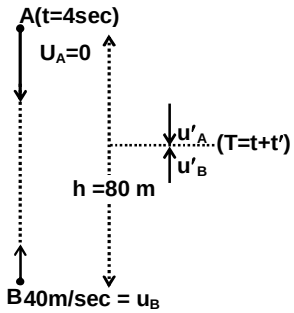
$\Rightarrow T = \rho ghr.$

26. $\frac{-GM}{2R} + \frac{G\frac{M}{4}}{\frac{5R}{2}} = -\frac{2GM}{5R}$

27. The apparent frequency initially decreases & then increases in one time period.

28. $\Delta E = -\frac{GM_e m}{12R} + \frac{GM_e m}{4R} = \frac{GM_e m}{6R}$

29. $v^2 = u^2 + 2gh \Rightarrow 0^2 = (40^2) - 2(10)(h)$



$\Rightarrow h = 80m$

where h is maximum height reached by ball A

$v = u - gt \Rightarrow t = 4$ sec,

$t' = \frac{80}{u_A + u_B} = 2\sec$

$u'_A = gt' = 20m/s$

$u'_B = 40 - 2 \times 10 = 20m/s$

Thus the speed of each ball gets interchanged (for equal mass & elastic collision)

30. $T_1 - T_2 = m\omega^2 r_1$

$T_2 = m\omega^2 (r_1 + r_2)$

$\Rightarrow T_1 = m\omega^2 (2r_1 + r_2)$

Thus $T_1 > T_2$



Chemistry

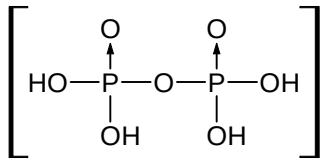
PART – II

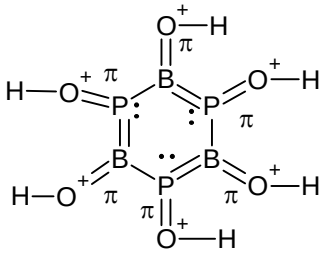
1. M_{wt} C_6H_5Cl = 112.5
M_{wt} $C_6H_5CH_3$ = 92
Given:
$x_{C_6H_5Cl} = 0.5$
So mass% composition reaction of C_6H_5Cl is
% $C_6H_5Cl = \frac{112.5}{204.5} \times 100 = 55\%$

2. As density increases, volume of gas decreases so intermolecular forces increase and diffisubility decreases.

3. For $\Psi_{432} \Rightarrow$
n = 4, l = 3, m = 2
Angular momentum = $\frac{h}{2\pi}\sqrt{\ell(\ell+1)} = \frac{\sqrt{12}h}{2\pi} = \frac{\sqrt{3}h}{\pi}$
Angular node = ℓ = 3
So total node = $\frac{\sqrt{3}h}{\pi} + 3 = \frac{\sqrt{3}h + 3\pi}{\pi}$

4. $2H_3PO_4 \xrightarrow{250^oC} \underset{\text{Pyrophosphoric acid}}{H_4P_2O_7} + H_2O$

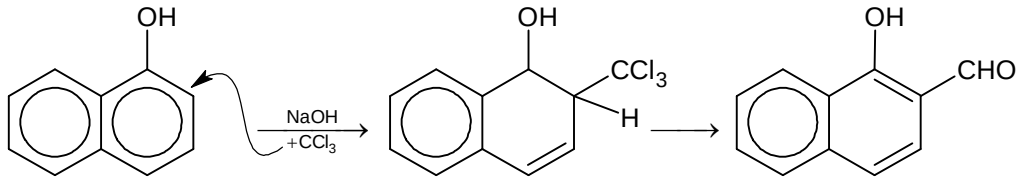


5. 

3 pπ—pπ back bonding between B and O and
3 pπ—dπ back bonding between P and O is possible.

7. $K_2Cr_2O_7 + 2H_2SO_4 \longrightarrow \underset{\text{Chromic anhydride (Red Crystal)}}{2CrO_3} + 2KHSO_4 + H_2O$

9. NH_2^- behaves as a base and E_2 elimination takes place.

10. 

11. $Na_2S_2O_3 + AgNO_3 \longrightarrow \underset{(white)}{Ag_2S_2O_3} \longrightarrow \underset{(black\ ppt)}{Ag_2S}$

12. $T_c = \frac{8a}{27Rb}$ so $T_c \propto \frac{1}{b}$; and $b = 4 \times N_A \times V \Rightarrow b \propto r$

13. AB type structure in which Zn^{+2} is tetrahedrally surrounded by S^- ion, S^{-2} is present in half of tetradedral acid.

14. It is an extensive property.

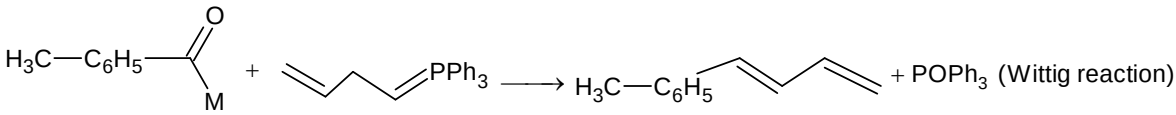
15. $\frac{d[B]}{dt} = [A]_0\left[\frac{K_1}{K_2 - K_1}\right]\left(-K_1e^{-K_1t} + K_2e^{-K_2t}\right)$

16. $ksp = \frac{\left[Fe(H_2O)OH\right]^{+2}\left[H_3O^+\right]}{\left[Fe(H_2O)_6\right]^{+3}}$, Ksp should be high.

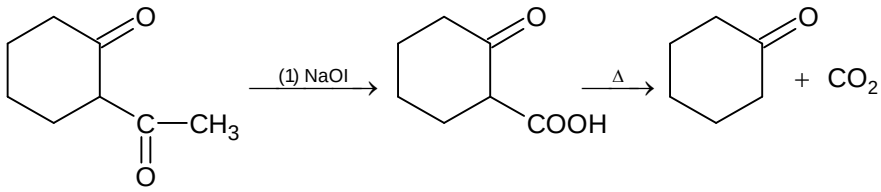
17. $\left[H^+\right] = \sqrt{K_{a_1}C_1 + K_{a_2C_2}} \Rightarrow 2.5$

18. ΔG^0 (net) = $\Delta G_1^0 + \Delta G_2^0$
Total e^- exchanged = (a + b)
$\Delta G^0 = -nFE^0$
$-(a+b)FE_3^0 = -aFE_1^0 - bFE_2^0$
$E_3^0 = \frac{aE_1^0 - bE_2^0}{(a-b)}$

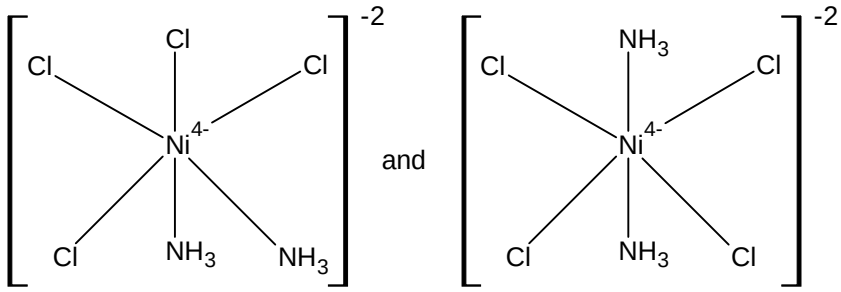
19. $K_2Cr_2O_7 \xrightarrow{\Delta} K_2CrO_4 + Cr_2O_3 + O_2$

20. 

21. Nitrogen is less electronegative than oxygen and on 3rd reactive site it is sp^3 hybridised.

22. 

23. As metallic character increases down the group so acidic nature of oxide decreases.

26. 

Cis Trans

27. $As_2O_3 + 5H_2O \longrightarrow 2AsO_4^{-3} + 10H^+ + 4e^-$

29. $2BeO + 2C \xrightarrow{\Delta} Be_2C + 2CO$

30. Only lithium nitride with formula LiN_3 in first group and Mg with Mg_3N.

Mathematics PART – III

2. Since the equations are consistent
$\therefore$ D = 0

$$\begin{vmatrix} (a+1)^3 & (a+2)^3 & -(a+3)^3 \\ (a+1) & (a+2) & -(a+3) \\ 1 & 1 & -1 \end{vmatrix} = 0$$

Put u = (a + 1), v = a + 2, w = a + 3
u – v = –1, v – wz –1, w – u = 2
u + v + w = 3a + 6

$$\therefore \begin{vmatrix} u^3 & v^3 & w^3 \\ u & v & w \\ 1 & 1 & 1 \end{vmatrix} = 0$$

(u – v) (v – w) (w – u) (u + v + w) = 0
(–1) (–1) (2) (3a + 6) = 0
i.e. a = –2.

3. 1234 = 3n + 1, so 1234^{567} = 3m + 1,89 = 3k – 1, so 89^{1011} = 3p – 1.
Hence $1234^{567} + 89^{1011}$ = 3n 1234 is even,
So 1234^{567} = 4q. 89 = 4r + 1, so 89^{1011} = 4t + 1.
Hence $1234^{567} + 89^{1011}$ = 4l + 1. So it is 9x + 1.

4. Here, $\tan A + \tan B = \frac{c^2}{ab}$ and $\tan A \cdot \tan B = 1$

$\tan A \cdot \tan B = 1 \Rightarrow \tan A = \cot B \Rightarrow A = 90^0 - B \Rightarrow A + B = 90^0 \Rightarrow C = 90^0$

$\Rightarrow \sin A = \frac{a}{c}$, $\sin B = \frac{b}{c}$

$\therefore \sin^2 A + \sin^2 B + \sin^2 C = \frac{a^2}{c^2} + \frac{b^2}{c^2} + 1 = 2$

5. $I_1 = \int_{-100}^{101} \frac{dx}{(5 + 2x - 2x^2)(1 + e^{2-4x})} = \int_{-100}^{101} \frac{dx}{(5 + 2(1-x) - 2(1-x)^2)(1 + e^{2-4(1-x)})}$

$\Rightarrow 2I_1 = \int_{-100}^{101} \frac{dx}{5 + 2x - 2x^2} = I_2$

$\Rightarrow \frac{I_1}{I_2} = \frac{1}{2}$.

6. Since f(x) and g(x) are one-one and onto and are also the mirror images of each other with respect to the line y = 2. It clearly indicates that h(x) = f(x) + g(x) will be a constant function and will always be equal to 4.

7. Equation of the curve passing through the points A, B, C, D is
$(3x + 4y - 24)(4x + 3y - 24) - \lambda xy = 0$
If it represents a parabola then the quadratic terms must form a perfect square.
$\Rightarrow \lambda = 1$ or 49.

8. Locus is a hyperbola with eccentricity = $\frac{r_1 + r_2}{r_1 - r_2} = \sqrt{2}$

9. In an equilateral triangle $r = \frac{R}{2}$

Also exradius $r_1 = 4R \sin\frac{A}{2} \cos\frac{B}{2} \cos\frac{C}{2} = 4R \cdot \frac{1}{2} \cdot \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} = \frac{3}{2}R$

$\Rightarrow$ r, R, r_1 are in A.P.

10. $\lim_{x\to\infty} \int_0^{x^2/2} \frac{t^2}{x^2(1+t^2)} dt = \frac{\infty}{\infty}$ form

$\Rightarrow \lim_{x\to\infty} \frac{\int_0^{x^2/2} \frac{t^2}{(1+t^2)} dt}{x^2} = \lim_{x\to\infty} \frac{\frac{x^4}{4+x^4} x}{2x} = \lim_{x\to\infty} \frac{x^5}{2x^5 + 8} = \frac{1}{2}$.

11. $(3m + 1)(3n - 1) = 3n^2 - 1$, so if p divides $3n - 1$, then it also divides $3n^2 - 1$
and hence also $(3n^2 - 1) - n(3n - 1) = n - 1$, and hence also $(3n - 1) - 3(n - 1) = 2$.
So $3n - 1 = 0, \pm 1$ or ± 2.
But n must be an integer, so 0, 1, – 2 do not work. – 1 gives n = 0 and hence m = 0, which is a solution. 2 gives n = 1 and hence m = 0, which is a solution.

12. Two lines can meet in a point
$\therefore {}^6C_2 = 15$
line and circle meet in two points = $({}^6C_1 \times {}^4C_1) \times 2 = 48$.
Two circles meet in 2 points = ${}^4C_2 \times 2 = 12$
total number of points = 48 + 15 + 12 = 75.

13. $1 + \cos 3x + 1 - \left[\cos\left(\frac{\pi}{2} + \left(2x - \frac{7\pi}{6}\right)\right)\right] = 0$

$2\cos^2\frac{3x}{2} + 1 - \cos\left(2x - \frac{2\pi}{3}\right) = 0$

$2\cos^2\frac{3x}{2} + 2\sin^2\left(x - \frac{\pi}{3}\right) = 0$

$\cos\frac{3x}{2} = 0,\ \sin\left(x - \frac{\pi}{3}\right) = 0$

$x = \frac{\pi}{3}, \pi$ and $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{7\pi}{3}, \ldots..$

$\therefore x = \frac{\pi}{3}$ is the common value which satisfies both

$\therefore x = 2n\pi + \frac{\pi}{3} = (6n + 1)\frac{\pi}{3}$

15. $\int \frac{(x^2+1)}{(x^4 - x^2 + 1)\cot^{-1}\left(x - \frac{1}{x}\right)} dx = \int \frac{\left(1 + \frac{1}{x^2}\right)}{\left(x^2 + \frac{1}{x^2} - 2 + 1\right)\cot^{-1}\left(x - \frac{1}{x}\right)} dx$

$= \int \frac{dt}{(t^2+1)\cot^{-1} t}$

Put $x - \frac{1}{x} = t \Rightarrow \left(1 + \frac{1}{x^2}\right)dx = dt$

$= \int \frac{-du}{u}$

Let $\cot^{-1} t = u$

$\Rightarrow -\frac{1}{(1+t^2)} dt = du = -\ln|u| + c$

$= -\ln\left|\cot^{-1}\left(x - \frac{1}{x}\right)\right| + c$

16. $\left[\frac{70}{3}\right] + \left[\frac{70}{9}\right] + \left[\frac{70}{27}\right] = 23 + 7 + 2 = 32$

17. Let λ be the period of sinx + cosax then
$\sin(\lambda + x) + \cos a(\lambda + x) = \sin x + \cos ax$ for all x
In this identity, putting x = 0 and $x = -\lambda$, we het
$\sin\lambda + \cos a\lambda = 1$ and $1 = -\sin\lambda + \cos\lambda$
solving these equation, we get
$\sin\lambda = 0$ and $\cos a\lambda = 1$
hence $\lambda = n\pi$ and $a\lambda = 2m\pi$. Where m, n are non–zero integers
hence $\frac{a\lambda}{\lambda} = \frac{2m\pi}{n\pi}$
or $a = \frac{2m}{n}$ (since $\lambda \neq 0$)

18. $A^2 = 0$, $A^3 = A^4 = \ldots = A^n = 0$
then $A(I \pm A)^n = A(I + nA)$
$= A \pm nA^2 = A$.

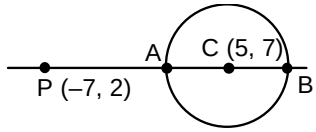
19. $\Delta = \sin^2 A + \sin^2(135° - A) = 1 + \sin^2 A - \sin^2(45° - A)$
$= 1 + \sin 45° \times \sin(2A - 45°) = 1 + \frac{1}{\sqrt{2}}.\sin\left(2A - 45^0\right)$
$= \Delta < \frac{\sqrt{2}+1}{\sqrt{2}}$ and $\Delta > \frac{1}{2}$
$0 < A < 135°$
$0 < 2A < 270°$
$-45° < 2A - 45° < 225°$
$-\frac{1}{\sqrt{2}} < \sin\left(2A - 45^0\right) < 1$
$\Rightarrow \frac{1}{2} < \sin^2 A + \sin^2 B < \frac{1+\sqrt{2}}{\sqrt{2}}$.

20. The curve xy = c and the circle $x^2 + y^2 = 1$ touches each other so
$x^2 + \frac{c^2}{x^2} - 1 = 0 \Rightarrow x^4 - x^2 + c^2 = 0$
will have equal roots so $(-1)^2 - 4c^2 = 0 \Rightarrow 4c^2 = 1$
$\Rightarrow c^2 = \frac{1}{4} \Rightarrow c = \frac{1}{2}$.
Roots of the equations $x^4 - x^2 + \frac{1}{4} = 0$ are $x = \pm \frac{1}{\sqrt{2}}$
$\Rightarrow y = \mp \frac{1}{\sqrt{2}}$.
Clearly distance between the points of contact = 2 units

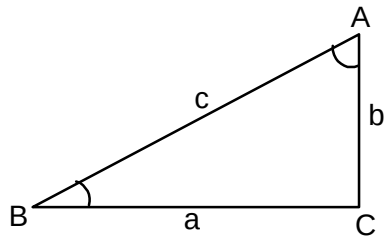
21. Sum of 7 digits = a multiple of 9
Sum of number 1, 2, 3, 4, 5, 6, 7, 8, 9 = 45 so two left should also have sum of 9.
The pair of left numbers is (1, 8), (2, 7), (3, 6), (4, 5).
with each pair left number of 7 digit number = 7!
So with all 4 pairs = $4 \times 7!$.

22. Equation of Ist line is $y - 0 = 1(x - 1)$
$y = x - 1$
and equation of IInd line is $x + y = 1$.
Clearly point of intersection is (1, 0).

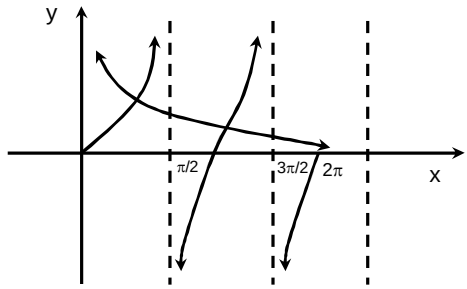
23. The centre c of the circle = (5, 7) and the radius
$= \sqrt{5^2 + 7^2 + 51} = 5\sqrt{5}$
$PC = \sqrt{12^2 + 5^2} = 13$
$\Rightarrow q = PA = 13 - 5\sqrt{5}$ and $p = PB = 13 + 5\sqrt{3}$
$\therefore$ G.M of p and q = $\sqrt{pq} = \sqrt{\left(13 - 5\sqrt{5}\right)\left(13 + 5\sqrt{5}\right)}$
$= \sqrt{169 - 125} = 2\sqrt{11}$.



24. a, b, c are in G.P $\Rightarrow b^2 = ac$
and $c^2 = a^2 + b^2$
$\Rightarrow c^2 = a^2 + ac$
$\Rightarrow \left(\frac{a}{c}\right)^2 + \left(\frac{a}{c}\right) - 1 = 0$
$\Rightarrow \frac{a}{c} = \cos B = \frac{-1 \pm \sqrt{5}}{2}$
$\Rightarrow \cos B = \frac{\sqrt{5} - 1}{2}$ (B is acute)
$\cos A = \sin B = \sqrt{1 - \cos^2 B} = \frac{\sqrt{\sqrt{5} - 1}}{2}$



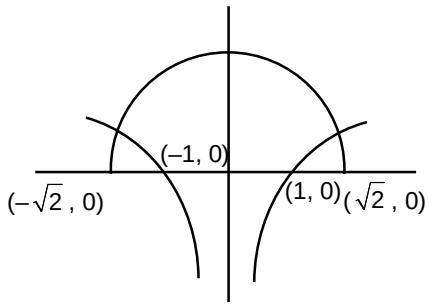
25. $y = \tan x = \frac{1}{x^2}$
From graph it is clear that it will have two real roots,



26. The given equation is $px^3 + qx^2 - rx - r = 0$. Its roots are given α, β, γ so we have
$\alpha + \beta + \gamma = \frac{-q}{p}$; $\alpha\beta + \beta\gamma + \gamma\alpha = -\frac{r}{p}$; $\alpha\beta\gamma = \frac{r}{p}$.
$D = \alpha\beta\gamma\left(1 + \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}\right) = \alpha\beta\gamma\left(\frac{\alpha\beta\gamma + \alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma}\right) = 0.$

27. For local maxima at $x \in R^+$ and minima at certain $x \in R^-$ the roots of $f'(x) = 0$ are of opposite sign
$\Rightarrow f'(0) < 0 \Rightarrow b < 0$.

28. Drawing the curves $y = \log_e |x|$
and $y = \sqrt{2 - x^2}$
So, number of solutions = 2



29. Putting $cx = c^2 + z$

$$c\int_{1+c}^{a+c} f(cx)dx = \int_{c}^{ac} f(c^2+z)dz$$

So, the required value = $c\int_{1+c}^{a+c} dx = c(a-1)$

30. $|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_1|^2$
$= 2(|z_1|^2 + |z_2|^2 + |z_3|^2) - (z_1\bar{z}_2 + \bar{z}_1z_2 + z_2\bar{z}_3 + \bar{z}_2z_3 + z_3\bar{z}_1 + \bar{z}_3z_1)$
$= 6 - (z_1\bar{z}_2 + \bar{z}_1z_2 + z_2\bar{z}_3 + \bar{z}_2z_3 + z_3\bar{z}_1 + \bar{z}_3z_1)$ .....(1)
Now $|z_1 + z_2 + z_3|^2 \geq 0$
$\Rightarrow |z_1|^2 + |z_2|^2 + |z_3|^2 + z_1\bar{z}_2 + \bar{z}_1z_2 + z_2\bar{z}_3 + \bar{z}_2z_3 + z_3\bar{z}_1 + \bar{z}_3z_1 \geq 0$
$\Rightarrow -(z_1\bar{z}_2 + \bar{z}_1z_2 + z_2\bar{z}_3 + \bar{z}_2z_3 + z_3\bar{z}_1 + \bar{z}_3z_1) \leq 3$ .....(2)
from (1) and (2)
maximum value = 6 + 3 = 9